Mastery in Data Analytics and Visualizations and Career Advancement

Lab sheet 08

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import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (
    confusion_matrix, accuracy_score, precision_score,
    recall_score, f1_score, ConfusionMatrixDisplay
)

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

# Convert TotalCharges to numeric and handle missing values
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df = df.dropna()

# Encode categorical variables
X = df.drop(columns=["customerID", "Churn"])
X = pd.get_dummies(X, drop_first=True)
y = LabelEncoder().fit_transform(df["Churn"])

X_train, X_test, y_train, y_test = train_test_split(
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    X, y, test_size=0.20, random_state=42, stratify=y
)

model = DecisionTreeClassifier(
    criterion="gini", max_depth=5, random_state=42
)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :", recall_score(y_test, y_pred))
print("F1 Score :", f1_score(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

ConfusionMatrixDisplay(cm).plot()
plt.title("Confusion Matrix")
plt.show()

plt.figure(figsize=(20, 10))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=["No Churn", "Churn"],
    filled=True,
```

```
    max_depth=3
)
plt.title("Decision Tree")
plt.show()

importance = pd.Series(model.feature_importances_, index=X.columns)
print("\nTop influential features:")
print(importance.sort_values(ascending=False).head(10))

# Questions and Answers
# Q1. What is classification, and how does it differ from regression?
# Classification predicts discrete class labels, while regression predicts continuous numerical values.
# Q2. Explain the working principle of the Decision Tree algorithm.
# A Decision Tree recursively splits the data using feature-based rules chosen to make the resulting groups more homogeneous with respect to the target class. Splitting continues until stopping criteria are reached, and a leaf provides the predicted class.
# Q3. Differentiate between Gini Index and Entropy as splitting criteria.
# Both measure impurity. Gini Index is based on the probability of misclassification, while Entropy measures information uncertainty. Lower impurity indicates a purer split.
# Q4. What is a Confusion Matrix? Explain its components.
# A confusion matrix summarizes classification results using True Positives, True Negatives, False Positives and False Negatives.
# Q5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?
# Accuracy is the proportion of all correct predictions. Precision is the proportion of predicted positives that are correct. Recall is the proportion of actual positives detected. F1-Score is the harmonic mean of precision and recall. Together they provide a more complete view than accuracy alone.
# Q6. What is overfitting in a Decision Tree? How can it be reduced?
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Overfitting occurs when the tree becomes too complex and models noise in the training data. It can be reduced using max_depth, min_samples_split, min_samples_leaf, pruning, and validation.

Q7. Why is customer churn prediction important for businesses?

Early identification of customers likely to leave enables targeted retention actions, improves customer satisfaction and can reduce revenue loss.

Q8. What are the advantages and limitations of the Decision Tree algorithm?

Advantages include interpretability, simple rule-based decisions, little need for feature scaling, and support for nonlinear relationships. Limitations include overfitting, instability to small data changes, and potentially poorer generalization than ensembles.

Q9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

Credit-risk assessment, medical diagnosis, and fraud detection are three examples.

Q10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

Organizations can identify high-risk customers, understand the features associated with churn, prioritize retention campaigns, personalize offers and measure which interventions reduce churn, thereby protecting revenue.

OUTPUT

PS D:\PROGRAMS\Career advancement lab\lab sheet 8> python -u "d:\PROGRAMS\Career advancement lab\lab sheet 8\Experiment_8_code.py"

Accuracy : 0.715

Precision: 0.6535433070866141

Recall : 0.8645833333333334

F1 Score : 0.7443946188340808

Confusion Matrix:

[[60 44]

[13 83]]

Top influential features:

Contract_Two year 0.279331

Contract_One year 0.271344

MonthlyCharges 0.135814

TotalCharges 0.128923

InternetService_Fiber optic 0.098994

tenure 0.061140

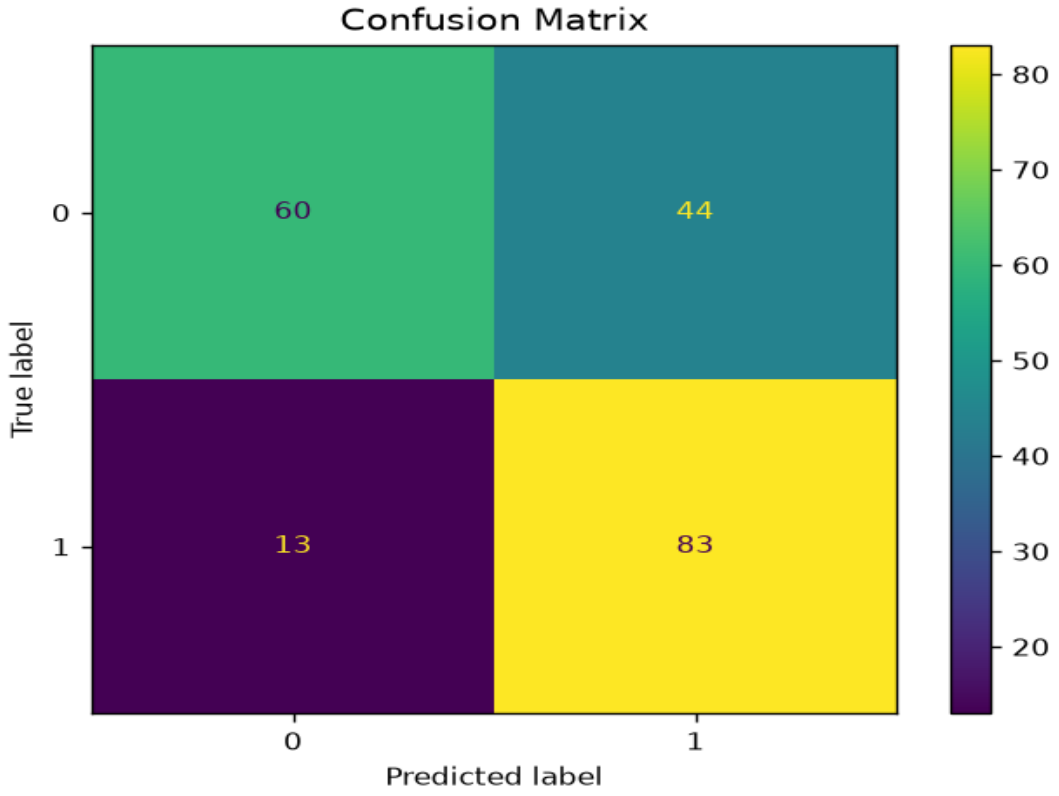
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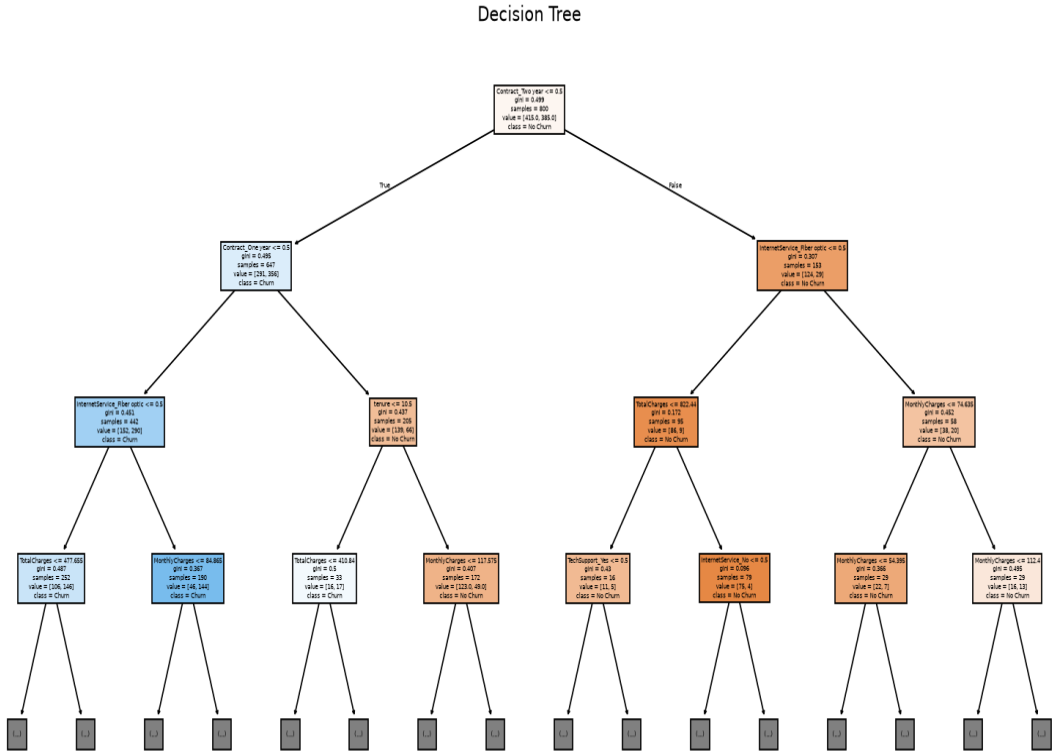
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Confusion Matrix
60
44
13
83
0
1
True label
0
1
Predicted label
80
70
60
50
40
30
20


Decision Tree